

AquaSense: An Explainable Machine Learning-Based Web Application for Water Quality Prediction Using pH, Temperature, and Turbidity

Bachelor of Science in Computer Science and Engineering

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[June, 2026]

Declaration

We hereby declare that this report titled *AquaSense: An Explainable Machine Learning-Based Web Application for Water Quality Prediction Using pH, Temperature, and Turbidity* is the result of work carried out by our project group for academic purposes. All sources of information, datasets, tools, libraries, and references used in this work have been acknowledged appropriately. The implementation, analysis, and documentation presented in this report are submitted as part of the requirements of the academic project/thesis.

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Abstract

Water quality monitoring is essential for public health, environmental protection, and sustainable water resource management. Conventional laboratory-based testing methods can be accurate but are often time-consuming, costly, and less suitable for rapid preliminary assessment. This study presents AquaSense, a lightweight and explainable machine learning-based web application for classifying water quality using three commonly measurable parameters: pH, temperature, and turbidity. A publicly available water pollution dataset was used for model development. The original five-class labels were mapped into three categories: Clean, Moderate, and Polluted. A threshold-informed scoring scheme was also applied using the selected parameters, and only records with agreement between the derived label and the mapped original label were retained to reduce label noise. Feature engineering was performed by deriving pH deviation, logarithmic turbidity, and temperature stress features. A Random Forest classifier was trained using the processed dataset, with SMOTE applied to the training data to address class imbalance. SHAP was used to provide interpretable explanations by showing the contribution of each parameter to individual predictions. The trained model was deployed as a Flask-based Docker application on Hugging Face Spaces. The system demonstrates how low-cost sensor-readable parameters can support preliminary, explainable water quality screening. Future work may extend the system using additional chemical, biological, and real-time sensor data.

Acknowledgements

First and foremost, we would like to express our sincere gratitude to **Abdul Kadir Ibne Kamal, Ph.D** for guidance, feedback, and support throughout this project. We are also thankful to the Department of Computer Science and Engineering, Bangladesh University of Professionals, for providing the academic environment necessary to complete this work.

We would like to acknowledge the contributions of all group members involved in different parts of the project, including data preparation, methodology development, model training, result analysis, discussion, documentation, and web deployment. Each member's contribution helped shape AquaSense into a complete machine learning-based water quality prediction system.

We are also grateful to the creators of the dataset and the developers of the open-source tools and libraries used in this project, including Python, scikit-learn, imbalanced-learn, SHAP, Flask, Docker, Matplotlib, and Hugging Face Spaces.

Finally, we would like to thank our families, friends, classmates, and well-wishers for their encouragement and support throughout the completion of this project.

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List of Abbreviations

AI	Artificial Intelligence
BOD	Biological Oxygen Demand
COD	Chemical Oxygen Demand
DO	Dissolved Oxygen
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
ML	Machine Learning
NTU	Nephelometric Turbidity Unit
pH	Potential of Hydrogen
RF	Random Forest
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
UI	User Interface
WHO	World Health Organization

Chapter 1

Introduction

1.1 Background

Water is one of the most essential natural resources for human life, agriculture, industry, and ecological balance. However, rapid urbanization, industrial discharge, agricultural runoff, and improper waste management have contributed to the degradation of water quality in many regions. Polluted water may contain physical, chemical, and biological contaminants that can cause health risks and environmental damage. As a result, water quality monitoring has become an important task for public health management and environmental protection.

Traditional water quality assessment often depends on laboratory testing, where samples are collected from water bodies and analyzed using specialized equipment. Although laboratory testing can provide accurate and detailed results, it may require trained personnel, time, and financial resources. In many practical scenarios, especially for preliminary field screening, there is a need for a faster and more accessible system that can provide an initial estimate of water quality using easily measurable parameters.

Recent developments in machine learning have made it possible to build predictive systems that classify or estimate water quality based on selected input parameters. Machine learning models can identify patterns in historical datasets and apply those patterns to new readings. However, for practical deployment, such systems should not only provide predictions but also explain the reasoning behind them. Explainability is important because users need to understand whether a prediction was influenced mainly by pH, turbidity, temperature, or other factors.

1.2 Problem Statement

Many water quality monitoring systems rely on multiple laboratory-based indicators such as dissolved oxygen, nitrate, ammonia, heavy metals, biological oxygen demand, chemical oxygen demand, and microbial contamination. While these indicators are important, they

may not always be available in low-cost field-level monitoring systems. A lightweight system using common sensor-readable parameters can help provide preliminary screening where complete laboratory testing is unavailable or delayed.

The main problem addressed in this project is the development of an accessible and explainable system that can classify water quality into meaningful categories using only pH, temperature, and turbidity. The system should be simple enough for non-specialist users while still using a trained machine learning model and interpretable explanations.

1.3 Objectives

The primary objective of this project is to develop AquaSense, an explainable machine learning-based web application for preliminary water quality prediction.

The specific objectives are:

1. To prepare a water quality dataset using selected sensor-readable parameters: pH, temperature, and turbidity.
2. To convert the original multi-class water pollution labels into a simplified three-class scheme: Clean, Moderate, and Polluted.
3. To apply feature engineering techniques that improve the representation of the selected parameters.
4. To use SMOTE for handling class imbalance in the training data.
5. To train a Random Forest classifier for water quality classification.
6. To deploy the trained model as a web application using Flask, Docker, and Hugging Face Spaces.

1.4 Scope of the Study

This study focuses on preliminary water quality classification using three selected parameters: pH, temperature, and turbidity. The system classifies water quality into three categories: Clean, Moderate, and Polluted. The project includes data preprocessing, label transformation, feature engineering, model training, explainability, and web deployment.

The scope does not include complete laboratory-grade water quality assessment. Important parameters such as dissolved oxygen, nitrate, ammonia, heavy metals, COD, BOD, and microbial contamination are not used in the final deployed model. Therefore, the system should be considered a preliminary screening tool rather than a replacement for full water quality testing.

1.5 Significance of the Study

The significance of AquaSense lies in its lightweight and explainable design. By using only pH, temperature, and turbidity, the system is compatible with low-cost sensor-based monitoring scenarios. The use of SHAP explainability also makes the system more transparent by showing which input factor influenced a particular prediction. Furthermore, deployment on Hugging Face Spaces makes the application accessible through a web browser without requiring users to install software locally.

1.6 Project Output

The final output of this project is a deployed web application that accepts pH, temperature, and turbidity values and returns:

- predicted water quality class,
- confidence score,
- class probability values,
- SHAP-based visual explanation,
- natural-language explanation of the dominant influencing factor.

The deployed application is available at:

<https://huggingface.co/spaces/Afsana24/aquasense>

Chapter 2

Literature Review

2.1 Water Quality Monitoring

Water quality monitoring is an important process for assessing the suitability of water for drinking, agriculture, industrial use, and ecological preservation. Commonly monitored parameters include pH, turbidity, temperature, dissolved oxygen, conductivity, biological oxygen demand, chemical oxygen demand, nutrients, heavy metals, and microbial indicators. Traditional Water Quality Index (WQI) computation relies on labor-intensive, multi-parameter laboratory processes, which limits rapid field assessments. Among these, pH, turbidity, and temperature are often measurable using relatively simple sensors and can provide useful preliminary information about water condition. Recent research demonstrated that Automated Machine Learning (AutoML) can achieve high water-quality prediction accuracy while requiring fewer input parameters than traditional approaches [1].

The pH value indicates the acidity or alkalinity of water. Extreme pH values may affect aquatic life and may also indicate contamination or chemical imbalance. Turbidity measures the cloudiness of water caused by suspended particles. High turbidity can reduce light penetration, affect aquatic ecosystems, and may indicate the presence of sediments, organic matter, or microbial contamination. Temperature influences chemical reactions, biological activity, dissolved oxygen levels, and microbial growth. For this reason, these three parameters can serve as useful indicators in a lightweight water quality screening system.

2.2 Machine Learning in Water Quality Prediction

Water quality monitoring traditionally relies on laboratory testing of parameters like total nitrogen, which is time-consuming and costly [2]. Recent advancements in machine learning allow for the rapid and low-cost prediction of water quality using easily accessible sensor data, such as pH, temperature, and conductivity [2]. Instead of manually defining all decision rules, machine learning models learn relationships between input parameters and

target quality labels from data. Models such as Decision Trees, Random Forests, Support Vector Machines, Logistic Regression, Artificial Neural Networks, and Gradient Boosting methods have been used in different water quality classification and prediction tasks.

Machine learning is useful in water quality prediction because environmental data often contain nonlinear relationships. For example, turbidity and pH may independently affect classification, but their combined effect may also influence water quality. Ensemble models such as Random Forest are particularly suitable because they can capture nonlinear patterns and interactions between features while remaining relatively robust.

2.3 Random Forest Classifier

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their outputs to produce a final prediction. For classification tasks, the class predicted by the majority of trees is selected as the final output. Random Forest reduces the overfitting risk associated with a single decision tree by averaging across many trees trained on random subsets of data and features.

Xu et al. evaluated Random Forest against several machine learning algorithms for predicting water quality parameters from available sensor inputs, finding that it consistently achieved superior predictive accuracy [2]. In this project, Random Forest was selected because it performs well on tabular datasets, can handle nonlinear relationships, does not require extensive feature scaling, and supports multi-class classification. It also provides feature importance values, which are useful for understanding the relative contribution of input variables.

2.4 Class Imbalance and SMOTE

Class imbalance occurs when one or more target classes have significantly fewer samples than others. In such cases, a classifier may become biased toward the majority class and perform poorly on minority classes. This is a common issue in environmental datasets because certain pollution conditions may occur less frequently than others.

The Synthetic Minority Over-sampling Technique, known as SMOTE, is a commonly used method for addressing class imbalance. SMOTE generates synthetic samples for minority classes by interpolating between existing minority samples. Patel et al. applied SMOTE alongside a comparative evaluation of Random Forest, Gradient Boosting, and other classifiers on a water potability dataset, reporting a best classification accuracy of 81% for the balanced models [3]. In this project, SMOTE was applied only to the training set after the train-test split. The test set was not resampled, preserving the integrity of model evaluation.

2.5 Explainable Artificial Intelligence and SHAP

Explainable Artificial Intelligence focuses on making machine learning predictions understandable to users. In sensitive decision-support systems such as water quality assessment, users may not fully trust a prediction unless they can understand the reasoning behind it. Without interpretability, it is difficult to determine whether a classification is driven primarily by pH, temperature, or turbidity.

SHAP (SHapley Additive exPlanations) is an explainability method grounded in cooperative game theory. SHAP estimates how much each input feature contributes to a specific prediction. Choudhary et al. integrated SHAP into a stacked ensemble model for WQI forecasting, demonstrating that SHAP values effectively identified the most influential water quality parameters and provided both local and global interpretability across individual predictions [4]. In AquaSense, SHAP values are computed for the model features and then aggregated into user-facing categories so that the explanation remains understandable for non-technical users.

2.6 Web-Based Water Quality Decision Support

A web-based system makes predictive models accessible to users without requiring programming knowledge. By deploying the model as a web application, users can enter water parameter values and immediately receive predictions and explanations. Hugging Face Spaces provides a convenient platform for hosting machine learning applications, including Docker-based applications. AquaSense uses a Flask backend with a custom HTML, CSS, and JavaScript frontend to provide an interactive prediction interface.

2.7 Research Gap

Many existing water quality prediction systems require a large number of laboratory-dependent parameters, which limits their applicability for low-cost field screening [2]. While Campos et al. showed that AutoML can achieve accurate WQI prediction with a reduced set of available parameters, their framework does not target the specific combination of pH, turbidity, and temperature as a minimal sensor-deployable input set [1]. Studies addressing class imbalance in water potability prediction, such as Patel et al., have applied SMOTE effectively but rely on model-level explainability through LIME rather than feature-level SHAP-based explanations [3]. Choudhary et al. demonstrated the value of SHAP for interpreting WQI predictions, yet their system was not packaged as an accessible web application for non-technical end users [4].

AquaSense addresses this gap by restricting its inputs to only three highly accessible parameters (pH, temperature, and turbidity) while coupling a SMOTE-balanced Random

Forest classifier with SHAP-based explanations and deploying the complete system as an interactive web application, providing a transparent and immediate decision support tool for field water quality screening.

Chapter 3

Methodology

3.1 Research Design

This study follows an applied machine learning research design. The project begins with dataset selection and preprocessing, followed by label transformation, feature engineering, model training, explainability integration, and web deployment. The final system is evaluated using standard classification metrics and demonstrated through a deployed web interface.

3.2 Data Collection

3.2.1 Field Data Collection

To obtain representative data from the study area, water samples were collected from multiple locations along the shore of the Turag River in Dhaka, Bangladesh. A total of 50 sampling points were selected, and their geographic coordinates were recorded using GPS-based mapping tools. The sampling locations were distributed along the riverbank to capture spatial variations in water conditions within the study area.

Due to limited access to laboratory facilities and water quality testing equipment, direct measurement of physicochemical parameters was not feasible. Therefore, a remote sensing and Geographic Information System (GIS)-based approach was adopted to estimate environmental parameters associated with the selected sampling locations.

Google Earth Engine (GEE) was used as the primary geospatial analysis platform. Landsat 9 Collection 2 Level-2 imagery was acquired for the study area and filtered based on spatial coverage, acquisition date, and cloud cover. Satellite images with cloud cover below 20% were selected to improve data quality. Surface temperature values were extracted from the thermal infrared band (ST_B10) for each sampling point using GEE's spatial sampling functions. The extracted values were exported as CSV files for further analysis.

The use of remote sensing techniques enabled the acquisition of environmental information corresponding to the field sampling locations despite the absence of laboratory measurements. The collected geospatial data were subsequently used for exploratory analysis and comparison with the machine learning prediction framework developed in this study.

The Google Earth Engine scripts used for data extraction are provided in the Appendix to ensure reproducibility of the data collection process.

3.2.2 Dataset for model training:

The dataset used in this study was obtained from Kaggle. It contains water quality-related records with multiple physical, chemical, biological, and geographical attributes. The original dataset contains 3,000 records and 25 columns. The original target label, `Pollution_Level`, uses five classes: Excellent, Good, Moderate, Poor, and Very Poor.

Although the raw dataset contains many parameters, this project selected only three input parameters for model development:

- pH,
- water temperature in degrees Celsius,
- turbidity in NTU.

These features were selected because they are commonly measurable using field-level sensors and match the intended lightweight deployment scenario.

Table 3.1: Selected input parameters

Parameter	Unit	Purpose
pH	–	Indicates acidity or alkalinity of water
Temperature	°C	Indicates thermal condition of water
Turbidity	NTU	Indicates suspended particles or cloudiness

3.3 Label Transformation and Annotation

First, rows containing missing values were removed. Rather than relying on the original dataset labels, which used a five-class scheme (Excellent, Good, Moderate, Poor, Very Poor) and multiple features, ground-truth class labels were derived programmatically from established WHO drinking-water standards. Each of the three parameters (pH, turbidity, and temperature) was assigned a quality score of 0, 1, or 2 according to the thresholds presented in Table 3.1. For each parameter, if the measured value falls within the ideal range (Score 0), it indicates no concern; if it falls within the acceptable range (Score 1), it indicates

mild deviation; if it falls outside both ranges (Score 2), it indicates significant deviation from WHO guidelines. The three individual scores were summed to produce a composite total score ranging from 0 to 6. Samples with a total score of 0 were labelled Clean, scores of 1 or 2 were labelled Moderate, and scores of 3 or above were labelled Polluted. The original five-class labels were mapped to this three-class scheme such that Excellent and Good were considered as Clean; Moderate was kept as Moderate; Poor and Very Poor were considered as Polluted. The dataset was then filtered to retain only records where the WHO-derived label matched the mapped original label, ensuring that only unambiguous samples were used for training and yielding a high-confidence labelled subset of the full dataset. The Final Class distribution after annotation is reported in Table 3.2.

Table 3.2: WHO-based Parameter Scoring Thresholds

Parameter	Score-0	Score-1	Score-2
pH	6.5–8.5	6.2–8.8	< 6.2 or > 8.8
Turbidity (NTU)	< 4	4–10	> 10
Temperature (°C)	< 25	25–28	> 28

Table 3.3: Final Class Distribution after Annotation

Water Quality Class	Count	Percentage
Clean	847	82.8%
Moderate	152	14.9%
Polluted	24	2.3%
Total	1023	100%

3.3.1 Dataset Preprocessing

Inspection of the training split revealed a pronounced class imbalance across the three quality categories. To address this, the Synthetic Minority Over-sampling Technique (SMOTE) was applied exclusively to the training set prior to model fitting because imbalanced data can cause the classifier to favor the majority class and perform poorly on minority classes. SMOTE generates synthetic samples for minority classes by interpolating between existing instances, thereby creating a balanced training distribution. The test set was kept separate and was not resampled to preserve real-world evaluation integrity. The final training set class distribution is reported in Table 3.4.

3.3.2 Feature Engineering

Feature engineering was performed because raw parameters alone may not capture the underlying patterns that influence water quality classification. Three additional features were

Table 3.4: Final Training Set Class Distribution (Before vs After SMOTE)

Class	Before SMOTE	After SMOTE
Clean	677	677
Moderate	122	677
Polluted	19	677
Total	818	2,031

derived from the raw inputs:

- **pH Deviation:** The absolute difference from neutral pH (7.0), used to capture the magnitude of acidity or alkalinity regardless of direction.
- **Log-Turbidity:** The natural logarithm of turbidity plus one, used to reduce the right-skewed distribution of NTU values and minimize the influence of extreme outliers.
- **Temperature Stress:** An ordinal encoding based on four temperature bins, designed to reflect increasing physiological and microbiological stress on water quality.

The final feature set comprised six variables: pH, Temperature_C, Turbidity_NTU, pH_deviation, log_Turbidity, and Temp_stress.

3.3.3 Model Training

A Random Forest classifier was selected as the primary model because it is robust to different feature scales, resists overfitting through ensemble averaging, and natively supports multi-class classification without requiring one-vs-rest strategies. The model was configured with 300 decision trees to ensure sufficient ensemble diversity, square-root feature selection to reduce correlation between trees, a minimum of two samples per leaf node to prevent overfitting, and balanced class weights as an additional safeguard against residual imbalance.

Training was performed on the SMOTE-augmented training set, and model performance was evaluated on the held-out 20% test set using accuracy, weighted precision, weighted recall, weighted F1-score, and five-fold stratified cross-validation.

3.3.4 Model Explainability via SHAP

SHAP (SHapley Additive exPlanations) was used to explain why the model made a particular prediction by showing how much each input feature contributed to the final decision. This helps users understand whether pH, temperature, or turbidity was the main factor driving the water quality classification.

To enable this, a `TreeExplainer` was initialized using the trained Random Forest model, along with a background dataset of 300 randomly sampled instances from the

SMOTE-augmented training set to serve as a reference for Shapley value computation. For deployment, SHAP focuses only on the three main input features shown to users, thereby avoiding confusion from the intermediate engineered features used internally by the model.

3.4 Web Application Deployment

The trained model, label encoder, SHAP `TreeExplainer`, and feature metadata were serialized using `joblib` and deployed within a Flask web application, containerized via Docker, and hosted on Hugging Face Spaces. As shown in the user interface in Figure 3.1, the front-end features a responsive, card-based layout built with HTML, CSS, and JavaScript, including dedicated input fields for temperature, pH, and turbidity. The interface also displays a World Health Organization (WHO) scoring reference matrix to improve user understanding.

When a user submits values, the application executes an automated three-step end-to-end pipeline. First, a WHO label assignment evaluates the inputs against predefined thresholds to establish contextual baseline classes. Second, the JavaScript front-end sends an asynchronous POST request to the Flask back-end, which executes the feature engineering pipeline (pH deviation, log-turbidity, and temperature stress) and feeds the processed data into the SMOTE-balanced Random Forest model to generate predictions along with class probabilities. Third, a SHAP explainability step leverages the `TreeExplainer` to calculate feature contributions, aggregating the engineered metrics back to the three source parameters.

The front-end then dynamically updates the result section by displaying the predicted class, confidence percentage, probability bars, an intuitive SHAP bar-chart visualization, and a natural-language summary identifying the dominant factor driving the final verdict. This deployment pipeline supports field-based water quality assessment and is publicly accessible at:

<https://huggingface.co/spaces/Afsana24/aquasense>

 AquaSense

PredictStandardsMethod

Water Quality Intelligence

Enter three simple measurements and our model, which is trained on WHO guidelines, classifies water quality and explains which parameters drive the result.

WHO-aligned labelsRandom Forest ModelSHAP explainability

Enter Water Parameters

TEMPERATURE °C	PH LEVEL Range: 0 – 14	TURBIDITY NTU

Analyse Water Quality

Figure 3.1: Web Application

Chapter 4

Results

This section presents the performance evaluation of the water quality classification model. The model is assessed using accuracy, precision, recall, F1-score, and confusion matrix. Finally, the model is applied to a real-world dataset collected from the Turag River to demonstrate its practical utility.

4.1 Dataset Summary

The raw dataset contained 3,000 records and 25 columns. After feature selection, label transformation, and consistency filtering, the final processed dataset contained 1,023 records.

Table 4.1: Dataset summary

Item	Value
Raw dataset size	3,000 records
Raw dataset columns	25 columns
Selected input features	pH, temperature, turbidity
Final dataset size	1,023 records
Final target classes	Clean, Moderate, Polluted

4.2 Model Performance

The Random Forest classifier was evaluated on a held-out test set consisting of 205 samples. Table 4.2 presents the overall performance metrics, while Table 4.3 provides detailed class-wise evaluation results.

The model achieves an overall accuracy of 98.05%, indicating that it correctly classifies approximately 98 out of every 100 water samples. The weighted precision of 97.93%

Table 4.2: Model Performance Metrics

Metric	Value
Accuracy	98.05%
Precision (weighted)	97.93%
Recall (weighted)	98.05%
F1-Score (weighted)	97.97%
5-Fold CV Accuracy	99.31% $\pm$ 0.29%

demonstrates that the model’s predictions are highly reliable across all classes. Similarly, the weighted recall of 98.05% shows that the classifier successfully identifies the majority of actual class instances. The weighted F1-score of 97.97% reflects a strong balance between precision and recall. Furthermore, the 5-fold cross-validation accuracy of 99.31% $\pm$ 0.29% indicates consistent performance across different training and validation splits, suggesting strong generalization capability and minimal risk of overfitting.

Table 4.3: Class-Wise Performance Metrics

Class	Precision	Recall	F1-Score	Support
Clean	0.9942	1.0000	0.9971	170
Moderate	0.9333	0.9333	0.9333	30
Polluted	0.7500	0.6000	0.6667	5
Accuracy	–	–	0.9805	205
Macro Average	0.8925	0.8444	0.8657	205
Weighted Average	0.9793	0.9805	0.9797	205

The *Clean* class achieves near perfect scores with precision of 0.9942 and recall of 1.0000. This indicates that the model rarely misclassifies a clean sample as another class and never misses an actual clean sample. The *Moderate* class shows strong performance with 0.9333 for both precision and recall, misclassifying only two samples out of 30. The *Polluted* class demonstrates the lowest performance with precision of 0.7500 and recall of 0.6000. This lower performance is attributed to the limited representation of polluted samples in the original dataset, even after SMOTE balancing, as the model has fewer real patterns to learn from this class.

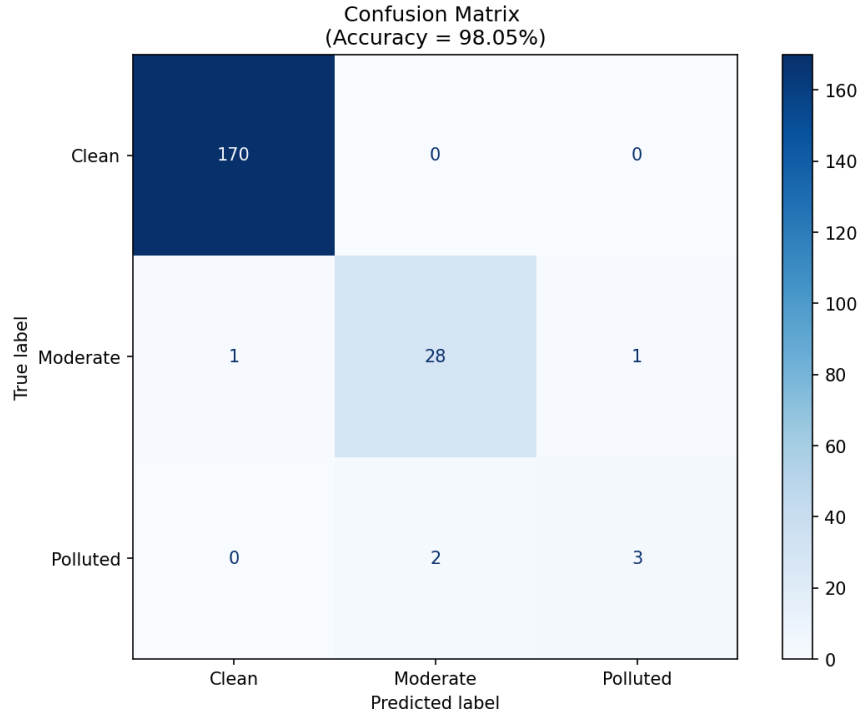


Figure 4.1: Confusion matrix of the Random Forest classifier

The confusion matrix reveals specific classification patterns across all three classes. For the Clean class, all 170 samples are correctly classified with no false positives or false negatives, indicating that clean water has distinct parameter signatures that the model reliably recognizes. For the Moderate class, 28 out of 30 samples are correctly identified. One Moderate sample is misclassified as Clean and one as Polluted, suggesting that moderate water occasionally exhibits parameter values near the boundaries of adjacent classes. For the Polluted class, only 3 out of 5 samples are correctly identified. The remaining 2 polluted samples are misclassified as Moderate, revealing systematic confusion between these two classes. This confusion is an expected outcome given the continuous nature of water quality parameters, where the distinction between Polluted and Moderate can be subtle, particularly when turbidity and pH deviation values approach WHO threshold boundaries.

4.3 Results on Real-World Dataset

The trained Random Forest model is applied to the real-world dataset provided by Shahed. This dataset consists of 51 water samples collected from the Turag River. Table 4.3 presents the prediction results for the first five samples.

All 51 samples are classified as Moderate water quality. Confidence scores range from 88.3% to 95.8%. No samples are classified as Clean or Polluted. This indicates that the water quality in the Turag River consistently falls within the Moderate category according to the model's learned WHO aligned criteria. This finding suggests that while the water does

Table 4.4: Prediction Results on Real-World Dataset (First 5 Samples)

Sample ID	pH	Temperature (°C)	Turbidity (NTU)	Predicted Class	Confidence (%)
P1	7.36	24.82	54.01	Moderate	93.8
P2	7.65	22.44	45.25	Moderate	91.3
P3	7.58	22.51	42.53	Moderate	95.8
P4	7.61	22.68	43.79	Moderate	94.5
P5	7.27	22.86	53.50	Moderate	65.8

not meet clean standards, it also does not reach the threshold for Polluted classification.

4.4 Web Application Testing

The deployed Hugging Face Spaces application was tested using several sample inputs. The system returned predictions, confidence scores, probabilities, and SHAP-based explanations.

Table 4.5: Example predictions from the deployed AquaSense web application

pH	Temperature	Turbidity	Prediction	Confidence
7.2	24°C	2.5 NTU	Clean	100.0%
7.2	24°C	50 NTU	Moderate	94.0%
4.5	32°C	100 NTU	Polluted	97.3%
9.2	27°C	8 NTU	Moderate	69.2%

The web application also validates user input. For example, when an invalid pH value above 14 was submitted, the system returned an error message stating that pH must be between 0 and 14.

Chapter 5

Discussion

5.1 Interpretation of Findings

The results indicate that a lightweight set of water quality parameters can be used to classify water condition into Clean, Moderate, and Polluted categories. The use of pH, temperature, and turbidity allows the system to remain compatible with simple field-level sensor readings. The Random Forest classifier is suitable for this task because it can learn nonlinear relationships between the input features and water quality labels.

The deployed examples show that the model responds logically to changes in input values. Low turbidity, near-neutral pH, and moderate temperature tend to produce a Clean prediction. Higher turbidity or greater deviation in pH and temperature tends to increase the probability of Moderate or Polluted classification.

5.2 Importance of Explainability

The integration of SHAP improves the transparency of the system. Instead of returning only a class label, AquaSense explains which parameter influenced the prediction most. This is important for practical use because users may need to know whether the water quality issue is mainly related to turbidity, pH, or temperature.

The aggregation of six internal features into three user-facing explanation groups makes the output easier to understand. A technical model may use engineered features such as `pH_deviation` and `log_Turbidity`, but a non-technical user is more likely to understand explanations based on pH, temperature, and turbidity.

5.3 Practical Significance

AquaSense can be useful as a preliminary screening tool for water quality assessment. Since the system is deployed as a web application, users can access it without installing

machine learning software. The system may be useful for educational demonstrations, early-stage environmental monitoring, and low-cost sensor-based water quality assessment.

5.4 Limitations

Despite its usefulness, the system has several limitations. First, it uses only three parameters: pH, temperature, and turbidity. Complete water quality assessment usually requires additional indicators such as dissolved oxygen, nitrate, nitrite, ammonia, phosphate, heavy metals, BOD, COD, and microbial contamination.

Second, the dataset used in this study was obtained from Kaggle rather than collected directly from local water bodies. Therefore, the model may not fully represent local environmental conditions. Third, the final training dataset was reduced from 3,000 records to 1,023 records after consistency filtering. Although this filtering was used to reduce label noise, it also reduced the amount of training data.

Finally, the system should not be interpreted as a replacement for laboratory testing. It is designed for preliminary screening and decision support rather than final regulatory assessment.

5.5 Unexpected Observations

The deployment test showed that the model can return very high confidence values for certain examples. While high confidence may indicate strong class separation, it should be interpreted carefully because machine learning confidence is not the same as laboratory certainty. Additional validation using real-world sensor data would be necessary before operational use.

Chapter 6

Conclusion and Recommendations

6.1 Conclusion

This project developed AquaSense, an explainable machine learning-based web application for preliminary water quality prediction. The system uses pH, temperature, and turbidity as input parameters and classifies water quality into Clean, Moderate, and Polluted categories. A public water pollution dataset was processed by selecting sensor-readable features, transforming the original five-class labels into a three-class structure, and filtering records based on agreement with a threshold-informed scoring scheme.

Feature engineering was applied to derive pH deviation, logarithmic turbidity, and temperature stress features. A Random Forest classifier was trained using the processed dataset, while SMOTE was used to address class imbalance in the training data. SHAP explainability was integrated to show how pH, temperature, and turbidity influenced individual predictions.

The trained model was deployed as a Flask-based Docker application on Hugging Face Spaces. Testing of the deployed application confirmed that it accepts user inputs, returns predictions with confidence scores, validates invalid inputs, and provides interpretable SHAP-based explanations. Overall, AquaSense demonstrates the potential of combining lightweight sensor parameters, machine learning, explainable AI, and web deployment for accessible preliminary water quality screening.

6.2 Recommendations

Based on the findings and limitations of this project, the following recommendations are proposed:

1. Future versions should include additional water quality parameters such as dissolved oxygen, nitrate, ammonia, COD, BOD, heavy metals, and microbial indicators.

2. The model should be validated using real-time sensor data collected from local water bodies.
3. More machine learning models should be compared, including Support Vector Machines, Gradient Boosting, XGBoost, and Neural Networks.
4. The dataset should be expanded to improve generalization across different geographic and environmental conditions.
5. A mobile or IoT-based version of the system may be developed for field deployment.
6. The web application should include data logging and historical trend visualization for long-term monitoring.

6.3 Future Work

Future work may focus on integrating AquaSense with physical sensors for real-time monitoring. Additional laboratory-based parameters can be included to improve prediction reliability. The system can also be extended with geolocation support, user authentication, cloud-based data storage, and dashboard visualization. With further validation, the project may serve as a foundation for an IoT-enabled environmental monitoring platform.

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Appendix A

Appendix A: Project Files

The local project directory contains the following important files:

- `es.ipynb`: notebook for dataset preparation, model training, evaluation, and export.
- `china_water_pollution_data.csv`: raw dataset.
- `dataset.csv`: processed dataset.
- `confusion_matrix.png`: confusion matrix figure.
- `feature_importance.png`: feature importance figure.
- `aquasense_model/rf_model.pkl`: trained Random Forest model.
- `aquasense_model/label_encoder.pkl`: label encoder.
- `aquasense_model/shap_explainer.pkl`: SHAP explainer.
- `aquasense_model/meta.json`: model metadata.

Appendix B

Appendix B: Deployment Link

The deployed AquaSense application is available at:

<https://huggingface.co/spaces/Afsana24/aquasense>

The runtime endpoint is:

<https://afsana24-aquasense.hf.space>

Appendix C

Appendix C: Sample API Request

The deployed application accepts prediction requests through a POST request to `/predict`. A sample JSON request is shown below:

```
{
  "ph": 7.2,
  "temp": 24,
  "turbidity": 2.5
}
```

A sample response includes the predicted class, confidence score, class probabilities, SHAP chart, SHAP explanation lines, and the top influencing factor.